PKM robot for respiratory simulation

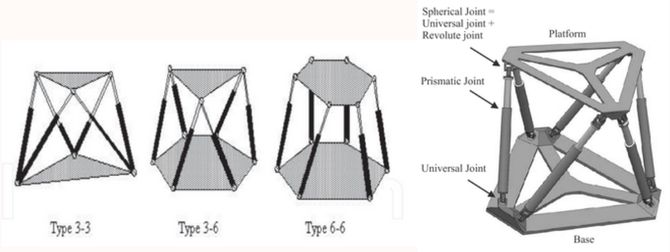
Shobit Verma

DESIGN PRESENTATION

Brief Background for PKM

- PKMs or parallel robots are the manipulators which having the end effector connected to the ground via several chains having actuations in parallel will have greater rigidity and superior positioning capability
- Used for applications where high load carrying capacity, good dynamic performance and precise positioning are of paramount importance, like flight simulator
- There are various configurations of this platform, but we would be looking at the 6 UPS (or 6 SPS) or commonly called the Stewart platform
- PKMs are particularly proposed for good ratios of strength-to-weight.
- the topology also suffers from its complex and small work-space, kinematic singularities and inconsistent dexterity which also limits its flexibility

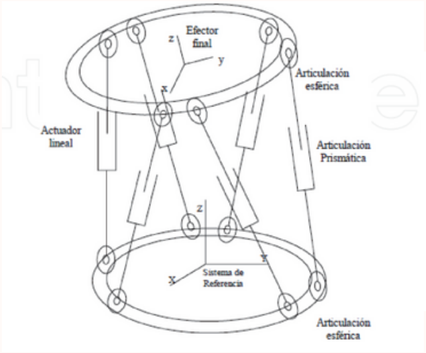
[1]



[2]

[3]

[4]



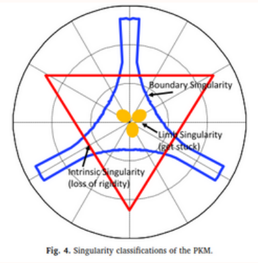


Fig. 4. Singularity classifications of the PKM.

REFERENCES

The Stewart platform manipulator: a review Bhaskar Dasguptaa,*, T.S. Mruthyunjaya [1]

Stewart-Gough Platform: Design and Construction with a Digital PID Controller Implementation Flabio Dario Mirelez-Delgado, José Ronaldo etc. [2]

Tlusty, J. Ziegert, S. Ridgeway, Fundamental comparison of the use of serial and parallel kinematics for machines tools, CIRP Ann. Manuf. Technol. 48 [3]

Optimization design for a compact redundant hybrid parallel kinematic machine Yen Ping-Lang a, Wang Ching-Henga, Lin Hao-Tingb, Hung Shuo-Sueic,d [4]

Existing works on this topic

- Until 2016, no studies have been done on simulating the respiratory motion via PKMs [1]
- After 2016, majority papers are for demonstrating chest compressions by PKMs [2]
- One such paper, has published study of the exact same task but in a scaled down version. [1]
- Regardless we have papers focusing on various aspects of the design such as -
    - Kinematics and statics [3]
    - Load capacity [4]
    - Control [5]
    - Overview [6]

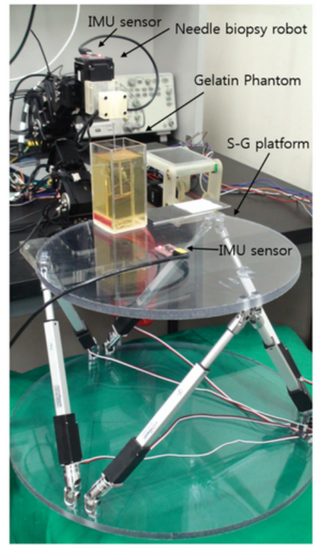
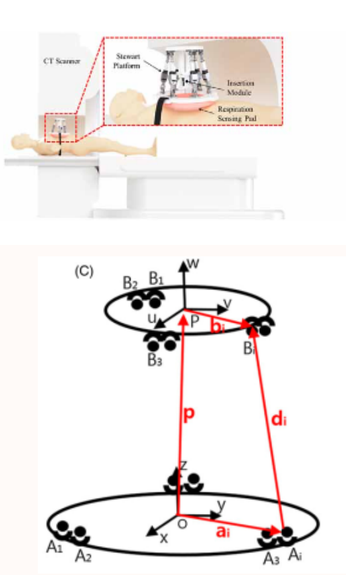


FIGURE 6 Test system (a) overview and (b) a gelatin phantom



REFERENCES

Impedance and admittance control for respiratory-motion compensation during robotic needle insertion – a preliminary test Yeoun Jae Kim | Jong Hyun Seo | Hong Rae Kim | Kwang Gi Kim * [1]

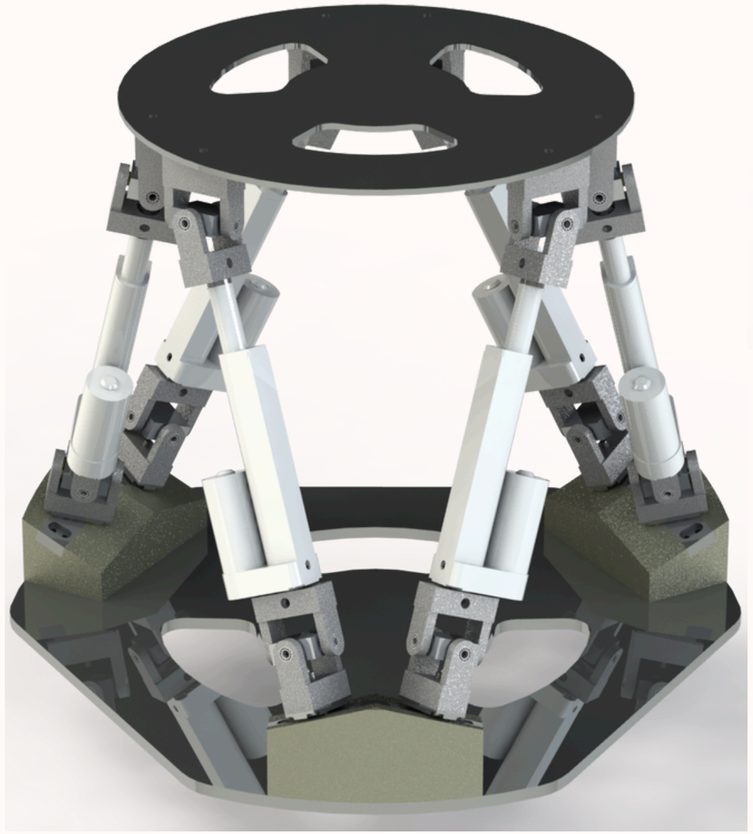
Autonomous Respiratory Motion Compensated Robot for CT-Guided Abdominal Radiofrequency Ablations : Anthony L. Gunderman; Mishek Musa; etc [2]

Kinematics and Statics of the Gough-Stewart Platform Jing-Shan Zhao * , Xiao-Cheng Sun and Song-Tao Wei [3]

Maximum Dynamic Load Carrying Capacity of a 6UPS-Stewart Platform Manipulator M.H. Korayem and M. Shokri1 [4]

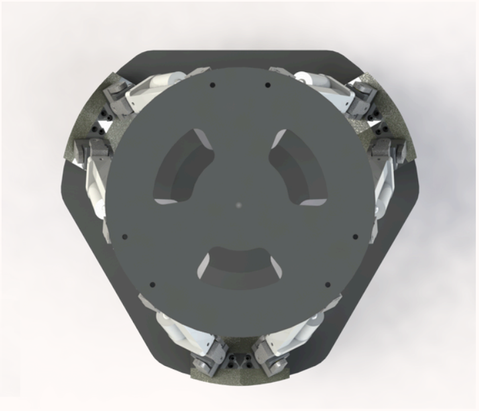
Stewart-Gough Platform: Design and Construction with a Digital PID Controller Implementation Flabio Dario Mirelez-Delgado, José Ronaldo etc [5]

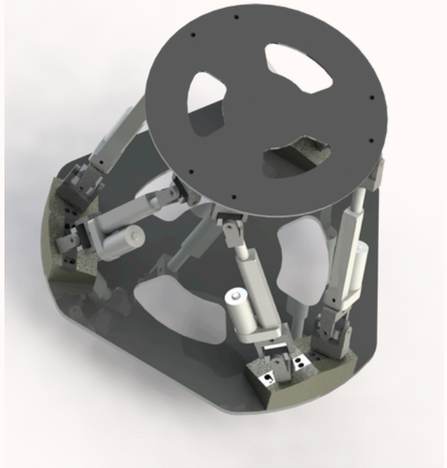
The Stewart platform manipulator: a review Bhaskar Dasgupta a,*, T.S. Mruthyunjaya b [6]

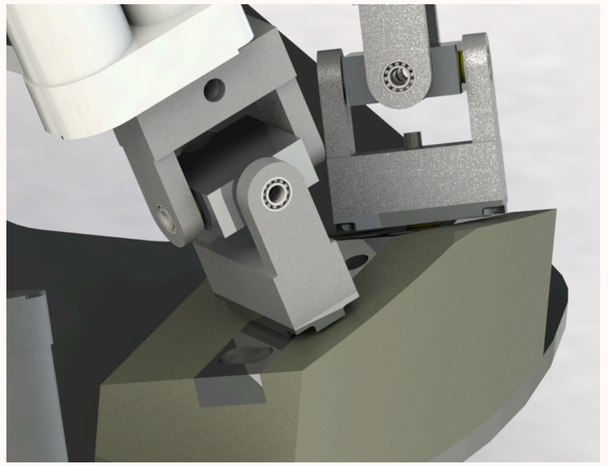


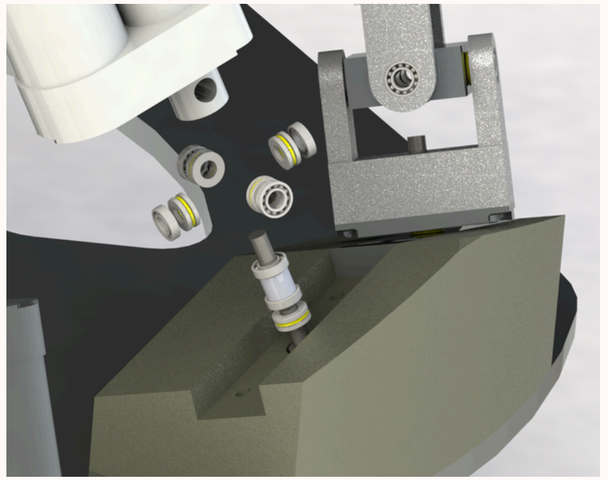




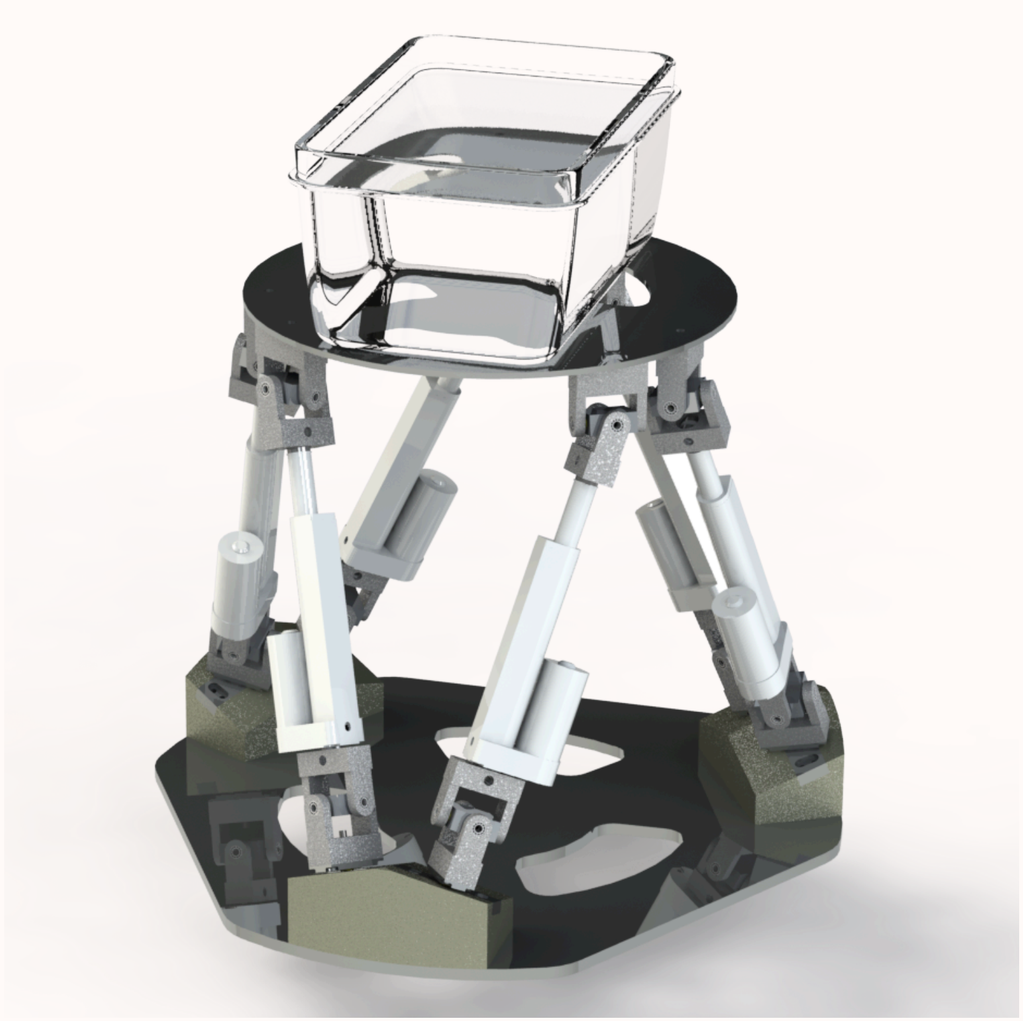


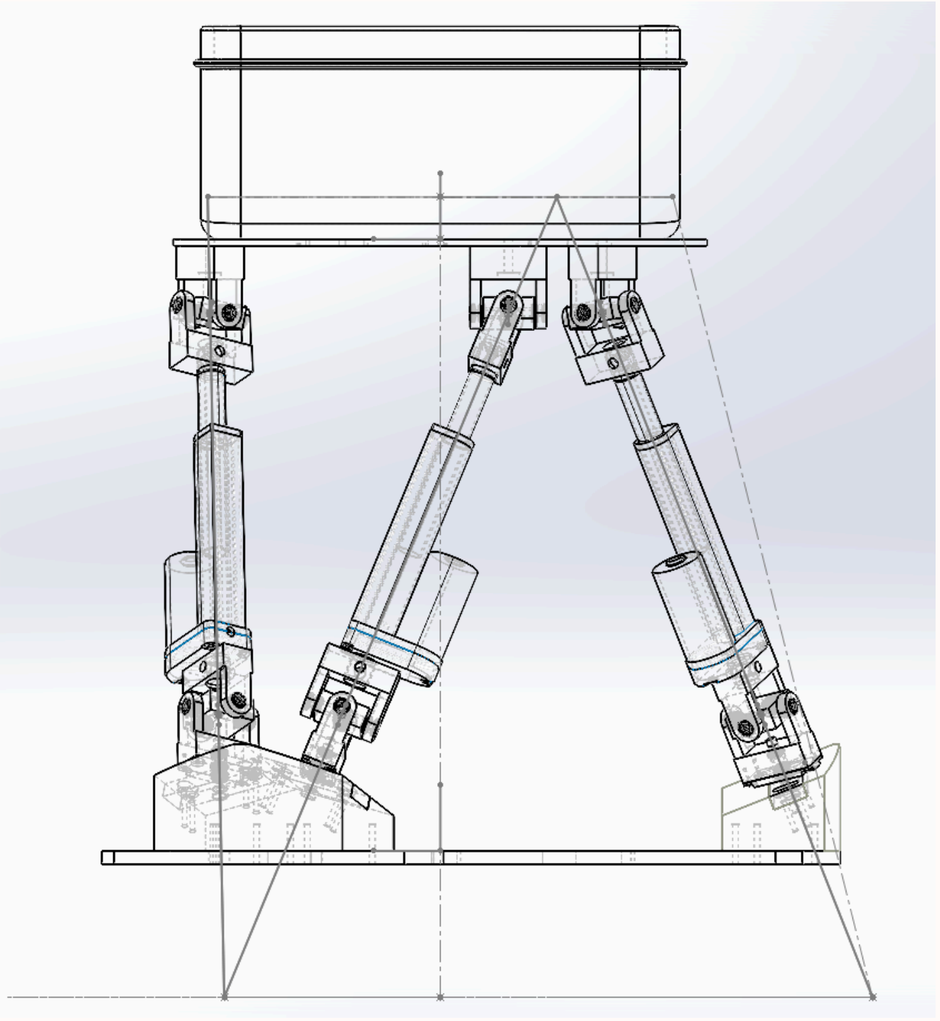






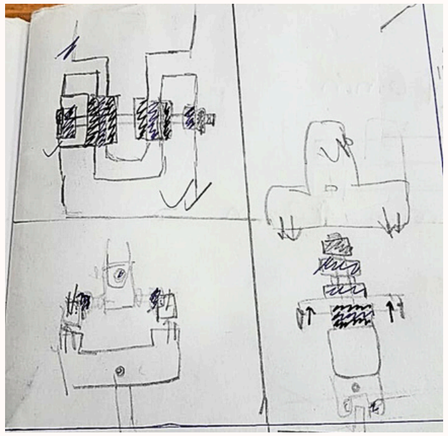
CAD MODEL

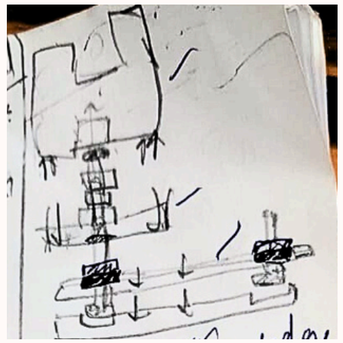


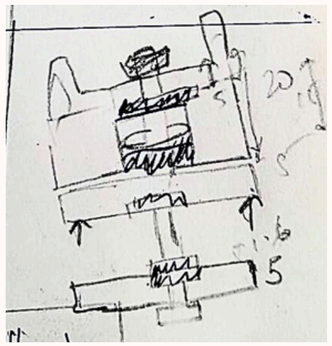


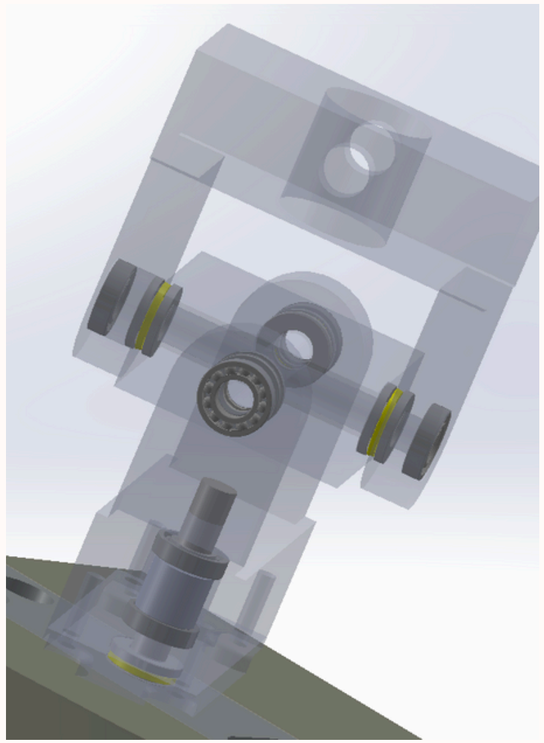
WITH PHANTOM

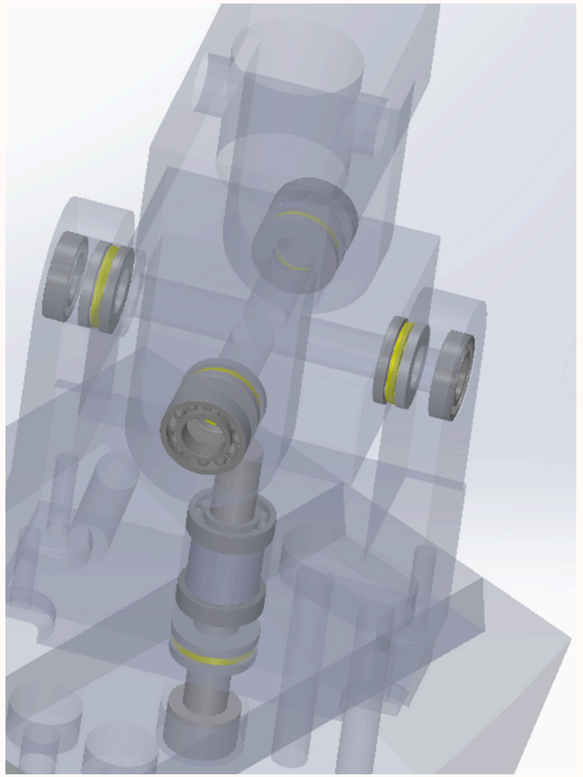
Self designed and assembled U-joints

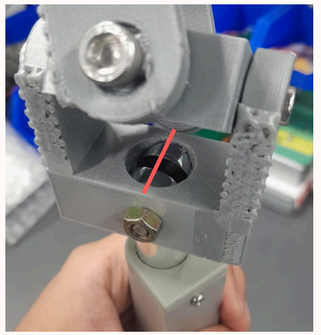




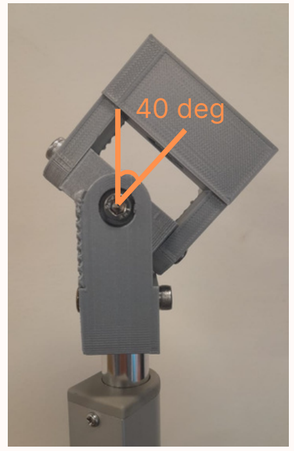








Joint might seem big but actually I have left little space only at the borders of the shaft. The shaft is thick

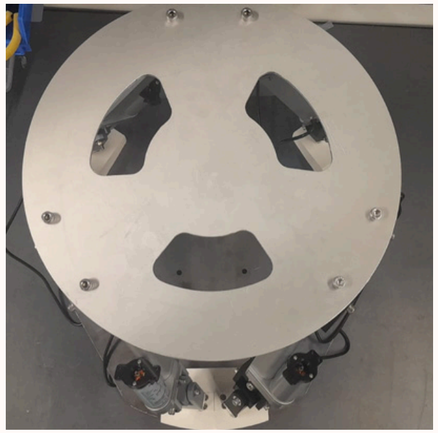


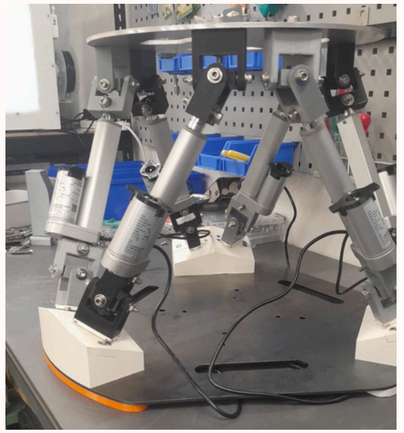
In total 108 bearings needed - 54 radial bearings(BR126) and 54 Thrust bearings (F6-12M)

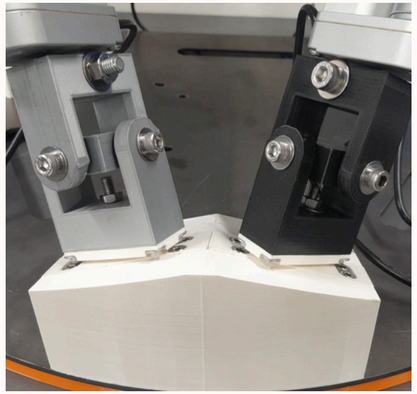
40 deg rotation is possible

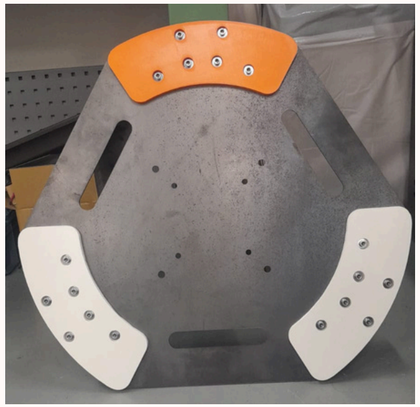


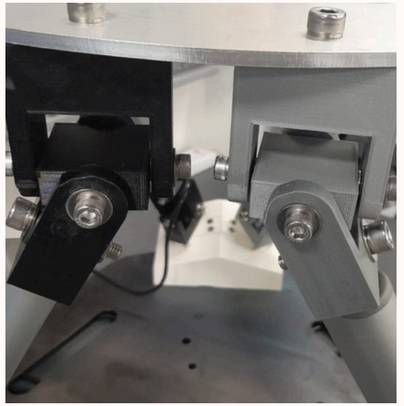












HARDWARE ASSEMBLED

Quantitatively

- Assembled **108** bearings
- Screwed **420** (258+162) times
- Lots of grinding (with machine)
- Lots of filing in and out
- Lots of gluing
- Assembled **341** (71+162+108) parts



THANK YOU!

